claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a74068f6b3fb34076.jsonl`
- SHA-256: `28b9415ab85d05ba2ea54711ccfd95b0515ca4c998b3b534d390a771d3ba52a6`
- Source modified: `2026-06-16T03:25:53+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf_3d536535-241`
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`

Transcript

1. user (2026-06-16T03:24:34.768Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question
Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):
- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).
- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.
- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RLOOCV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"On THUMOS14, TemporalMaxer achieves 67.7% average mAP (mAP@0.5=71.8, @0.7=44.7), beating the ActionFormer baseline (66.8% avg) while using 16.4 GMACs and only 7.1M parameters versus ActionFormer's 45.3 GMACs / 29.3M params ? i.e. a parameter-free temporal block roughly 4x smaller than the attention model still wins on accuracy."

Source: https://arxiv.org/abs/2303.09055 (primary)
Supporting quote: "ActionFormer[60] I3D[7] 82.1 77.8 71.0 59.4 43.9 66.8 80 ...
Our(TemporalMaxer) I3D[7] 82.8 78.9 71.8 60.5 44.7 67.7 50"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

refuted=true if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
refuted=false ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:24:38.419Z)

I'll verify this claim about TemporalMaxer vs ActionFormer on THUMOS14. Let me check the primary source.

3. assistant (2026-06-16T03:24:45.936Z)

I'll fetch the actual TemporalMaxer paper to verify the specific numbers claimed.

4. user (2026-06-16T03:24:50.315Z)

I cannot extract the specific THUMOS14 results table you're requesting. While the webpage content provided includes the abstract and metadata for the TemporalMaxer paper (arXiv:2303.09055), it does not contain the actual results tables, performance metrics, or computational resource comparisons you're asking about.

To obtain the exact numerical values for TemporalMaxer and ActionFormer's mAP scores at different thresholds (0.3, 0.4, 0.5, 0.6, 0.7), their average mAP, GMACs/FLOPs, and parameter counts, you would need to access the full PDF of the paper via the link provided: https://arxiv.org/pdf/2303.09055

5. assistant (2026-06-16T03:24:53.081Z)

The abstract page doesn't have the tables. Let me fetch the full PDF and also search for corroborating sources.

6. user (2026-06-16T03:24:59.013Z)

I apologize, but I cannot extract the THUMOS14 results table you're requesting from the provided content. The document appears to be a PDF with compressed/encoded stream data that isn't readable in plain text format.

To obtain the specific mAP values at different tIoU thresholds (0.3, 0.4, 0.5, 0.6, 0.7) for both TemporalMaxer and ActionFormer, along with their GMACs and parameter counts, you would need to:

1. Access the original paper in a readable format (HTML or properly extracted PDF text)
2. Locate the THUMOS14 benchmark results section
3. Reference the comparison table directly

If you have access to the paper's text version or can share a clearer excerpt of the results table, I'd be happy to help extract and compare those specific metrics.

[Binary content (application/pdf, 1.1MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579965172-5ov5x2.pdf]

7. user (2026-06-16T03:25:03.659Z)

Web search results for query: "TemporalMaxer THUMOS14 67.7 average mAP ActionFormer 66.8 parameters GMACs"

Links: [{"title":"TemporalMaxer: Maximize Temporal Context with only Max Pooling","url":"https://arxiv.org/pdf/2303.09055"},{"title":"TemporalMaxer Performance in the Face of Constraint: A Study in Temporal","url":"https://repository.tudelft.nl/file/File_7430e57c-b269-4010-a020-376bf7180e18"},{"title":"Benchmarking Data Efficiency and Computational Efficiency of Temporal   Action Localization Models","url":"https://arxiv.org/pdf/2308.13082"},{"title":"GitHub - happyharrycn/actionformer_release: Code release for ActionFormer (ECCV 2022) · GitHub","url":"https://github.com/happyharrycn/actionformer_release"},{"title":"Benchmarking Data Efficiency and Computational Efficiency","url":"https://cveu.github.io/2023/papers/11.pdf"},{"title":"[2202.07925] ActionFormer: Localizing Moments of Actions with Transformers","url":"https://arxiv.org/abs/2202.07925"},{"title":"GitHub - f1ibrahim-tmu/TemporalMaxer: TemporalMaxer: Maximize Temporal Context with only Max Pooling for Temporal Action Localization · GitHub","url":"https://github.com/f1ibrahim-tmu/TemporalMaxer"},{"title":"GitHub - TuanTNG/TemporalMaxer: TemporalMaxer: Maximize Temporal

Context with only Max Pooling for Temporal Action
Localization","url":"https://github.com/TuanTNG/TemporalMaxer"},{"title":"A Multi-Modal
Transformer Network for Action
Detection","url":"https://arxiv.org/pdf/2305.19624"},{"title":"Adaptive Perception Transformer
for Temporal Action Localization","url":"https://arxiv.org/pdf/2208.11908"}]

Based on the search results, I found relevant information about these models:

TemporalMaxer achieves an average mAP of 67.7 on THUMOS14 with a kernel size of 3, while ActionFormer reached an average mAP of 66.8% on the THUMOS'14 dataset.

Regarding computational efficiency, TemporalMaxer outperforms ActionFormer with 2.8x fewer GMACs and 3x faster inference speed. This indicates that TemporalMaxer achieves superior performance while being significantly more computationally efficient in terms of both parameters (GMACs measure computational operations) and inference time compared to ActionFormer.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. assistant (2026-06-16T03:25:06.213Z)

The WebFetch couldn't parse the PDF, but a copy was saved locally. Let me read it directly, and also check the GitHub README which typically reproduces the results table.

9. user (2026-06-16T03:25:09.049Z)

pdftoppm failed: Command 'pdftoppm' not found or is in an unsafe location (current directory)

10. user (2026-06-16T03:25:14.159Z)

THUMOS14 Results Analysis

Based on the README, here are the exact results from the THUMOS table:

TemporalMaxer's average mAP on THUMOS: "67.7"

However, the document does not provide GMACs or parameter count in the THUMOS results table. The README only shows IoU thresholds (0.3-0.7) and average performance.

ActionFormer comparison: The README does not include an ActionFormer results table for THUMOS. The only mention of ActionFormer appears in the introduction, which states: "TemporalMaxer outperforms the robust baseline, ActionFormer, with 2.8x fewer GMACs and 3x faster inference speed." However, no specific numerical values for ActionFormer's GMACs, parameters, or mAP are provided in the document.

11. assistant (2026-06-16T03:25:17.657Z)

The PDF reader needs poppler. Let me try reading the PDF text with a Python tool and verify the specific per-threshold numbers and the GMACs/params claim. The claim makes a critical assertion: "2.8x fewer GMACs" is what the paper says, but the claim states specific numbers "16.4 GMACs / 7.1M params vs 45.3 GMACs / 29.3M params" and "roughly 4x smaller." Let me dig into these specifics.

12. user (2026-06-16T03:25:19.734Z)

pypdf ok

13. user (2026-06-16T03:25:27.632Z)

```
==== KW: THUMOS ====
TAL on various challenging datasets including THUMOS
[22], EPIC-Kitchens 100 [11], MultiTHUMOS [57], and
```

across a variety of challenging datasets, namely THUMOS
[22], EPIC-Kitchens 100 [11], MultiTHUMOS [57], and
THUMOS14 and MultiTHUMOS datasets, roughly 20 min-
4.1. Results on THUMOS14
Dataset. THUMOS14 dataset [22] contains 200 valida-
features of THUMOS14 dataset using two-stream I3D [7]
the THUMOS14 dataset [22] with state-of-the-art meth-
THUMOS. It is 1.6x faster than the ActionFormer baseline
Table 1. The results obtained on the THUMOS14 dataset [22] are presented for various tIoU
thresholds, with the average mAP calculated
4.4. Results on MultiTHUMOS
Dataset. The MultiTHUMOS dataset [57] is a densely
labeled extension of THUMOS14, consisting of 413 sports
action categories per video, compared to THUMOS14. As
such, it poses a greater challenge for TAL than THUMOS.
While MultiTHUMOS are being used in action detection
of our model on MultiTHUMOS and compare it against the
lowing [47], to extract features for MultiTHUMOS. The
tiTHUMOS dataset. We report the results at different tIoU thresh-
mance on the MultiTHUMOS dataset [57] with recent state-
the MultiTHUMOS dataset, and the results [10, 48, 9] are
tion set of the THUMOS14 dataset.
Table 5. Ablation studies about different TCM blocks on THUMOS14. Inference times are measured
using an input video with 2304 clip
thumos challenge on action recognition for videos ?in the

==== KW: ActionFormer ====
Common Architecture AFSD ActionFormer Ours
layer in AFSD [28] to capture local temporal context, Graph [25] in G-TAD [54] and Transformers
[51] in ActionFormer [60] to model
in self-attention in ActionFormer [60] where the red boxes exhibit
ing the ActionFormer[60].
ing that ActionFormer [60] employs Transformer [51] as a
bust baseline model, ActionFormer [60]. This setup in-
Transformer block in ActionFormer with the Max Pooling
TAGS [38], GLFormer [19], ActionFormer [60], or Graph-
THUMOS. It is 1.6x faster than the ActionFormer baseline
and 8.0x faster than ActionFormer, respectively.
ActionFormer [60] I3D [7] 82.1 77.8 71.0 59.4 43.9 66.8 80
ActionFormer [60] + GAP [37] I3D [7] 82.3 ? 71.4 ? 44.2 66.9 >80
and robust baseline, ActionFormer [60], with an average
ActionFormer [60] at every tIoU threshold. Notably, Tem-
ActionFormer [60] 26.6 25.4 24.2 22.3 19.1 23.5
ActionFormer [60] 25.2 24.1 22.7 20.5 17.0 21.9
ActionFormer [60] 35.9 26.9 15.2 26.2
strong baseline ActionFormer [60].
ActionFormer [60] 46.4 32.4 15.0 28.6
vided by the authors to implement ActionFormer [60] on
decreases by 7.4% mAP compared with the ActionFormer
line, ActionFormer. To clarify, TemporalMaxer effectively
outperforms the robust baseline, ActionFormer, with 2.8x
than ActionFormer backbone [63], 20.1 ms.
[60] Chen-Lin Zhang, Jianxin Wu, and Yin Li. Actionformer: Lo-

==== KW: GMACs ====
TCM tIoU GMACs? #params (M)? time (ms)? backbone time (ms)?0.5 0.7 Avg.
fewer GMACs and 3x faster inference speed. Especially,

==== KW: Param ====
the extracted video clip features with a basic, parameter-
while requiring signi?cantly fewer parameters and com-
sive parameters and computations.
works have incorporated extensive parameters, high computational costs, and complex modules in a
backbone such as 1D Convolutional
from pretrained 3D CNN by only utilizing a basic, parameter-free, local operating Max Pooling

block. Our proposed method, the simplest
non-maximum suppression (NMS) hyper-parameters in the
tensive and highly parameterized attention module with the
TCM tIoU GMACs? #params (M)? time (ms)? backbone time (ms)?0.5 0.7 Avg.
quire much computational as well as parameters, and be
Secondly, the convolution layer introduces the most param-
eters which tend to overparameterize the model. We argue
parameters, which may lead to over?tting and thus less gen-
Transformer with a parameter-free operation, subsampling
prisingly, this none-parameter operation achieves higher re-
contain many parameters, and the features from pretrained
model ever for TAL task that contains minimalist param-
basic, non-parametric and temporally local operating max-
state-of-the-art methods with sophisticated, parametric and

==== KW: 67.7 ====
mance, achieving an average mAP of 67.7% mAP, out-
Our (TemporalMaxer) I3D [7] 82.8 78.9 71.8 60.5 44.7 67.7 50
TemporalMaxer 71.8 44.7 67.7 16.4 7.1 10.4 2.5
average mAPs are 67.7, 67.1, 66.8, 65.7, respectively. Our

==== KW: 66.8 ====
AMNet [35] I3D [7] 76.7 73.1 66.8 57.2 42.7 63.3 ?
ActionFormer [60] I3D [7] 82.1 77.8 71.0 59.4 43.9 66.8 80
Transformer [51] 71.0 43.9 66.8 45.3 29.3 30.5 20.1
average mAPs are 67.7, 67.1, 66.8, 65.7, respectively. Our

==== KW: 71.8 ====
Our (TemporalMaxer) I3D [7] 82.8 78.9 71.8 60.5 44.7 67.7 50
TemporalMaxer 71.8 44.7 67.7 16.4 7.1 10.4 2.5

==== KW: 44.7 ====
Our (TemporalMaxer) I3D [7] 82.8 78.9 71.8 60.5 44.7 67.7 50
TemporalMaxer 71.8 44.7 67.7 16.4 7.1 10.4 2.5

==== KW: 7.1 ====
BC-GNN [4] TSN [52] 57.1 49.1 40.4 31.2 23.1 40.2 ?
BMN [29, 11] 10.8 9.8 8.4 7.1 5.6 8.4
Conv [28] (Our Impl) 62.8 37.1 59.4 45.6 30.5 16.3 9.0
Subsampling 64.3 37.7 61.0 16.2 7.1 10.4 2.5
Average Pooling 66.1 39.4 63.2 16.4 7.1 10.4 2.5
TemporalMaxer 71.8 44.7 67.7 16.4 7.1 10.4 2.5
average mAPs are 67.7, 67.1, 66.8, 65.7, respectively. Our

==== KW: 29.3 ====
Transformer [51] 71.0 43.9 66.8 45.3 29.3 30.5 20.1

==== KW: 45.3 ====
Transformer [51] 71.0 43.9 66.8 45.3 29.3 30.5 20.1

==== KW: 16.4 ====
Average Pooling 66.1 39.4 63.2 16.4 7.1 10.4 2.5
TemporalMaxer 71.8 44.7 67.7 16.4 7.1 10.4 2.5

==== KW: 2.8 ====
TAGS [38] I3D [7] 68.6 63.8 57.0 46.3 31.8 52.8 ?
TadTR [35] I3D [7] 74.8 69.1 60.1 46.6 32.8 56.7 195*
Our (TemporalMaxer) I3D [7] 82.8 78.9 71.8 60.5 44.7 67.7 50
100 dataset, achieving an average mAP of 24.5% and 22.8%
Our (TemporalMaxer) 26.3 25.2 23.5 21.3 17.6 22.8
Conv [28] (Our Impl) 62.8 37.1 59.4 45.6 30.5 16.3 9.0
outperforms the robust baseline, ActionFormer, with 2.8x

14. assistant (2026-06-16T03:25:42.600Z)

I now have the exact tables from the paper. Let me verify every number in the claim against the extracted text.

Table 1 (THUMOS14 main results):
- `ActionFormer [60] I3D [7] 82.1 77.8 71.0 59.4 43.9 66.8 80` ? ActionFormer: @0.3=82.1, @0.4=77.8, @0.5=71.0, @0.6=59.4, @0.7=43.9, **avg=66.8**
- `Our (TemporalMaxer) I3D [7] 82.8 78.9 71.8 60.5 44.7 67.7 50` ? TemporalMaxer: @0.5=71.8, @0.7=44.7, **avg=67.7**

Table 5 (ablation, includes GMACs/params):
- `Transformer [51] 71.0 43.9 66.8 45.3 29.3 30.5 20.1` ? ActionFormer's Transformer block: @0.5=71.0, @0.7=43.9, avg=66.8, **GMACs=45.3, #params=29.3M**, time=30.5ms, backbone=20.1ms
- `TemporalMaxer 71.8 44.7 67.7 16.4 7.1 10.4 2.5` ? TemporalMaxer: @0.5=71.8, @0.7=44.7, avg=67.7, **GMACs=16.4, #params=7.1M**, time=10.4ms, backbone=2.5ms

Now let me verify each assertion in the claim:

Claim	Paper	Match?
TemporalMaxer avg mAP = 67.7%	67.7	?
mAP@0.5 = 71.8	71.8	?
mAP@0.7 = 44.7	44.7	?
ActionFormer avg = 66.8%	66.8	?
TemporalMaxer 16.4 GMACs	16.4	?
TemporalMaxer 7.1M params	7.1	?
ActionFormer 45.3 GMACs	45.3	?
ActionFormer 29.3M params	29.3	?

One nuance to check: the claim says "parameter-free temporal block roughly 4x smaller than the attention model." 29.3/7.1 = 4.13x on params; 45.3/16.4 = 2.76x on GMACs (paper says "2.8x fewer GMACs"). So "roughly 4x smaller" refers to parameters (4.1x), which is accurate. The paper's own headline is "2.8x fewer GMACs" but the claim's "4x" is explicitly tied to parameter count (29.3M vs 7.1M = 4.1x), which the paper's Table 5 supports.

Every number is directly verifiable in the paper's Table 1 and Table 5, confirmed against both the primary PDF and the supporting quote. The supporting quote even shows the exact Table 1 rows. The only minor framing point: GMACs/params come from Table 5 (the TCM-block ablation), not Table 1, but they are the paper's own reported figures for the full models, and the claim correctly distinguishes "GMACs" from "params" with the right ratios.

The claim is fully supported, current (2023 CVPRW paper, appropriate for a "recent techniques" survey), from a primary source (arXiv), and not cherry-picked ? it is the paper's headline comparison reproduced accurately.

15. user (2026-06-16T03:25:49.709Z)

Structured output provided successfully

16. assistant (2026-06-16T03:25:53.888Z)

Verified the claim. All eight numerical assertions (TemporalMaxer 67.7% avg / 71.8 @0.5 / 44.7 @0.7 / 16.4 GMACs / 7.1M params; ActionFormer 66.8% avg / 45.3 GMACs / 29.3M params) match the primary source exactly, drawn from Table 1 and Table 5 of arXiv:2303.09055. The "~4x smaller" framing correctly refers to parameters (4.13x), distinct from the paper's "2.8x fewer GMACs" headline, and the claim reports raw GMACs separately without conflating the two. Claim **not refuted**, high confidence.